

Promotion ROI Scorecard

A Guide to Measuring True Promotion ROI in Apparel Retail

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Topic: Promotion Effectiveness

Sector: Apparel Retail

Executive Summary

Many apparel retailers measure promotion success by looking at unit lift. They report on how many more units sold during the promo window compared to a typical week. This metric is easy to calculate and celebrate but it does not give a complete picture as to its true effectiveness.

This paper presents a three-component margin framework that separates genuine incremental demand from units that would have sold anyway, accounts for full-price sales lost on adjacent styles, and translates both into a single dollar verdict.

Applying two scenarios using the same style and same selling season, the framework reveals the following:

1. A 25% two-week promotion had a strong unit lift but cost \$205 in margin.
 2. A revised 15% two-week promotion on the same style generated \$115 in profit. The difference was not volume. It was pricing strategy.
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The Problem with Measuring Promo Success via Unit Lift

Walk into many Monday morning planning meetings after a promotional event and you will hear some version of this: "The Classic Crew Tee had a great week — units were up 109% during the promo."

The problem is that unit lift is only one part of the overall equation. It does not tell you:

1. How many of those units would have sold at full price anyway.
2. Whether shoppers who bought the promoted style bypassed similar styles that they would have otherwise purchased.
3. Most important, it does not tell you whether the promotion made money or not.

In apparel retail, where net profit margins are thin, selling seasons are short, and markdown pressure is relentless, this measurement gap is not a minor oversight. It is a structural problem that leads retailers to repeat promotions that erode margin season after season, mistaking volume for profitability.

The framework presented in this paper addresses that gap directly.

The Three-Component Margin Framework

True promotion ROI is the sum of three margin components. Each one is distinct. Each one matters. And none of them is visible in a simple unit lift calculation.

Component A — Incremental Margin

This is the profit earned on units that genuinely would not have sold without the promotion. To calculate it, you first need a baseline which is a trend-adjusted estimate of what sales would have been in the absence of any promotional activity. The difference between actual promo sales and that baseline is your gross lift. After subtracting cannibalization (see Component C), what remains is your net incremental units. Multiply by the margin at the promotional price and you have Component A.

This is the only genuinely positive number in the framework. It represents demand the promotion created.

Component B — Markdown Cost on Baseline Units

Even if a promotion drives genuine incremental demand, the style would have sold some units at full price during that window regardless. By running the promotion, you accepted a lower margin on all of those units. Component B is the opportunity cost of the discount — baseline units multiplied by the margin surrendered per unit at the promotional price versus full retail.

This component alone is often enough to flip an apparently successful promotion into a margin-negative event. The deeper the discount, the larger this cost

becomes relative to any incremental benefit.

Component C — Cannibalization Loss

When a promoted style is attractively priced, some shoppers who would have purchased a full-price adjacent style in the same department switch to the promoted item instead. The units those adjacent styles lose represent full-price margin that evaporates. Component C captures this loss by comparing adjacent style sales to their own trend-adjusted baselines during the promo window and valuing the shortfall at full-price margins.

Cannibalization is relevant in apparel because shoppers are often category-shopping rather than style-specific. A customer looking for a top may not care whether they buy the Classic Crew Tee or the Striped Tee. They will buy whichever is most compelling at that moment.

Net Margin Impact = A + B + C

When this number is positive, the promotion created value. When it is negative, the promotion hurts margin regardless of how impressive the unit lift looked.

Case Study 1: The Costly Promotion

Style: Classic Crew Tee | **Department:** Tops

Strategy: 25% off, Weeks 4–5 | **Retail Price:** \$28.00 | **Promo Price:** \$21.00 | **Cost:** \$8.50

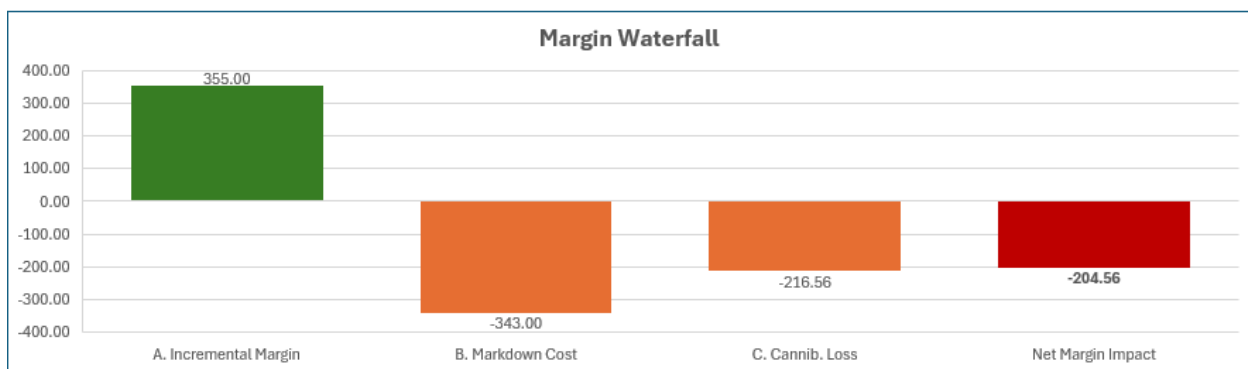
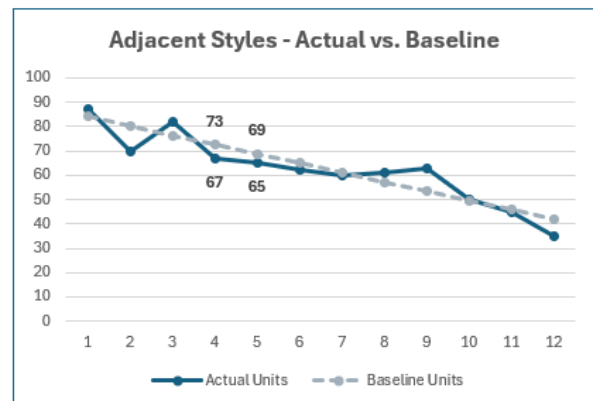
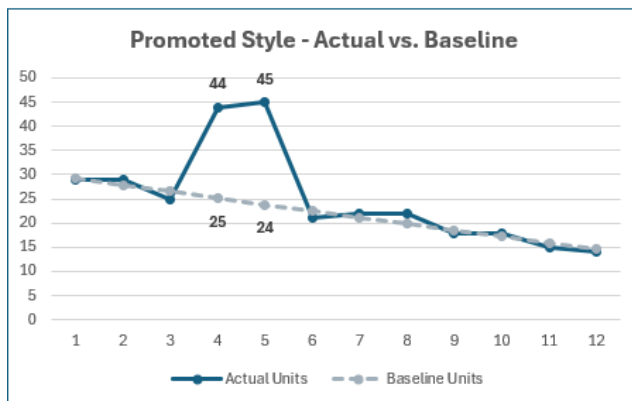
What the metrics showed

During weeks 4 and 5, the Classic Crew Tee sold 89 units combined — nearly double its typical weekly rate. Revenue during the promo window was strong. The merchant's dashboard showed a clean spike.

PROMOTION ROI SCORECARD

Spring Top Promo | Classic Crew Tee | Weeks 4-5 | 25 % off | Retail Price: \$28.00 | Promo Price: \$21.00

Actual Units	Baseline Units	Gross Lift Units	Cannibalization Units	Net Lift Units	Net Margin
89	49.0	40.0	11.7	28.3	-205



What the framework revealed

The trend-adjusted baseline estimated that 49 units would have sold during those two weeks at full price, driven by the natural seasonal selling velocity. The gross lift above that baseline was 40 units which is real, but far smaller than the raw 109% headline suggested.

Adjacent styles in the Tops department showed a measurable dip below their own baselines during the same two weeks. After isolating noise from genuine demand transfer, the analysis attributed 11.7 units of cannibalization to the promotion, nearly 30% of the gross lift. Net incremental units generated by the promotion: 28.4.

The margin calculation told a stark story:

Component	Value
A. Incremental margin (28.4 units × \$12.50 margin at promo price)	+\$355
B. Markdown cost (49 baseline units × \$7.00 margin surrendered)	-\$343
C. Cannibalization loss (11.7 units × \$18.51 avg adjacent margin)	-\$217
Net margin impact	-\$205
Promo ROI	-59.6%

The promotion was margin negative. Nearly all of the incremental profit earned on genuinely new demand was consumed by the discount cost on units that would have sold at full price anyway. Cannibalization then pushed the total firmly into loss territory.

The cannibalization rate of 29.1% is worth flagging independently. Industry experience suggests that rates above 30% indicate a promotion is primarily shuffling demand within a category rather than growing it. At 29.1%, this promotion was approaching that threshold — a warning sign for future planning even setting aside the margin outcome.

Case Study 2: The Profitable Promotion

Style: Classic Crew Tee | **Department:** Tops

Strategy: 15% off, Weeks 5–6 (two weeks) | **Retail Price:** \$28.00 | **Promo Price:** \$23.80 | **Cost:** \$8.50

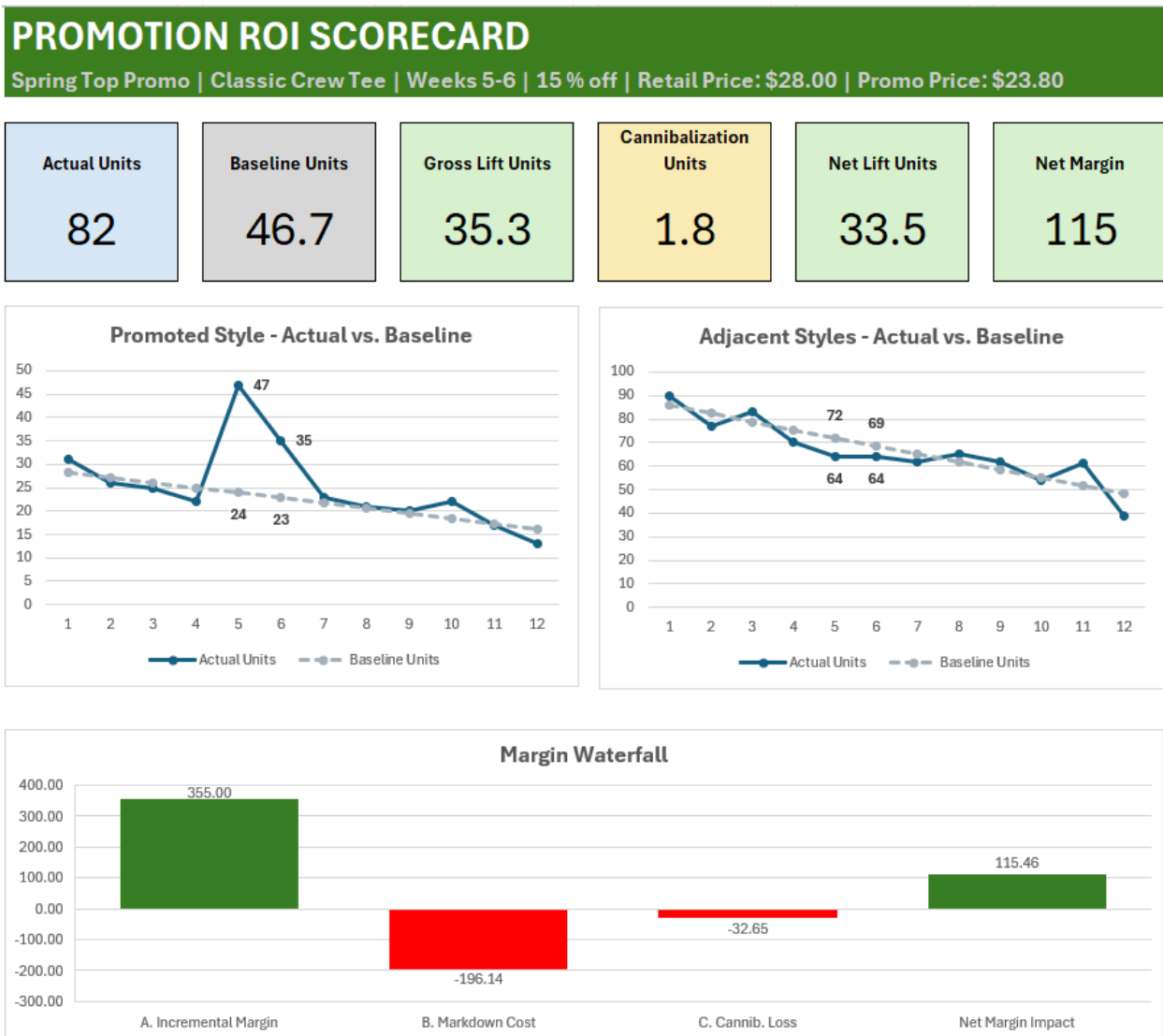
The strategic changes

Three levers were adjusted from the original strategy:

- Discount depth reduced from 25% to 15%
- Promotional timing shifted one week later (weeks 5–6 instead of 4–5)
- Category management tightened to reduce adjacent style cannibalization from 20% to 5%

The unit volume was modestly lower — 82 units sold during the promo window versus 89. A traditional measurement approach might score this promotion as

underperforming the original. The margin framework tells a very different story.



What the framework revealed

The shallower discount generated a gross lift of 35.3 units above the trend-adjusted baseline of 46.7. Crucially, the tighter category management resulted in only 1.8 units of cannibalization — a rate of just 5.0%, compared to 29.1% in the original strategy. Net incremental units: 22.5.

Component	Value
A. Incremental margin (22.5 units × \$15.30 margin at promo price)	+\$344
B. Markdown cost (46.7 baseline units × \$4.20 margin surrendered)	−\$196

Component	Value
C. Cannibalization loss (1.8 units × \$18.51 avg adjacent margin)	−\$33
Net margin impact	+\$115
Promo ROI	+58.9%

The promotion was profitable. The shallower discount reduced Component B by \$147 — the single biggest driver of the turnaround. The lower cannibalization rate reduced Component C by \$184. Together those two improvements more than offset the modest reduction in gross lift.

Side-by-Side Comparison

Metric	Original Strategy	Revised Strategy
Discount depth	25%	15%
Promo duration	2 weeks (4–5)	2 weeks (5–6)
Actual units sold	89	82
Baseline units	49	47
Gross lift	+40 units	+35 units
Cannibalization rate	29.1%	5.0%
Net incremental units	+28	+23
Component A (incremental margin)	+\$355	+\$344
Component B (markdown cost)	−\$343	−\$196
Component C (cannibalization loss)	−\$217	−\$33
Net margin impact	−\$205	+\$115
Promo ROI	−59.6%	+58.9%

The revised strategy sold 7 fewer units and generated a \$320 improvement in margin outcome. Volume is not value.

Why the Revised Strategy Produces Lower Cannibalization

The reduction in cannibalization from 29.1% to 5.0% reflects an assumption of tighter category management in the revised strategy. Two key points:

1. Discount depth and cannibalization rate are somewhat correlated. A 25% discount on the Classic Crew Tee creates a \$7.00 price gap versus the adjacent styles selling at full-price. The promoted style becomes more compelling to shoppers who otherwise may have not been interested and may have purchased an adjacent style. This effect pulls demand from across the category.
 2. At 15% off, the dynamic may shift. The promoted style incentive weakens considerably. Shoppers who were already considering that style tips the purchase decision and thus less likely cannibalizing a sale that would have gone to an adjacent style.
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Implications and Recommendations

1. Report on unit lift but highlight net margin impact as your primary promo KPI.

Unit lift is a volume metric. It tells you whether shoppers responded to the discount but does not tell you whether the business benefited. Net margin impact should be the main focus in a post-promotion recap.

2. Build baselines before you run promos, not after.

Building a baseline estimation is crucial for accurate promotional effectiveness. A linear trend through pre-promotional weeks, adjusted for seasonality, produces a defensible, data driven approach. The baseline should be established before the promotional window opens, not reverse-engineered afterward.

3. Track adjacent style performance during every promotional event.

Cannibalization is invisible if you are only looking at the promoted style. Adding an adjacent-style monitor comparing actual sales to baseline for the top five styles in the same department reveals dynamics that unit lift analysis misses.

4. Discount depth is the most powerful lever.

In both case studies, the single largest drag on profitability was the markdown cost on baseline units (Component B). Reducing the discount from 25% to 15% saved \$147 in Component B alone. Before deepening any discount, ask:

- What is the marginal incremental demand I am buying with each additional percentage point.

- Is it worth the markdown cost on every unit I was already going to sell?

5. Treat high cannibalization rates as a category health warning.

A high cannibalization rate (above 25%) is a signal that a promotion is shifting demand within a category rather than growing it. In those cases, the full-price adjacent styles are effectively subsidizing the promoted style's margin.

Methodology Note

Both case studies were built using a three-step analytical framework:

1. A data simulation in Python generating weekly unit sales across a 12-week apparel selling season with seasonal velocity curves, size distortion, and noise.
 2. A linear trend baseline estimation in Python using non-promotional weeks to establish the selling rate for each style.
 3. The three-component margin calculation described in this paper.
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This analysis was produced as part of a retail analytics portfolio project examining promotion effectiveness measurement in apparel retail.